Practical Workbook

CT-265

Introduction to Tools and Techniques for Data Science

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LAB # 01

OVERVIEW OF PYTHON

EXERCISE # 01: BASIC INPUT AND OUTPUT

Q1. Write a program that takes a student's name, roll number, and GPA as input and displays them clearly.

SOLUTION:

```python
name=input("Enter your Name:")
rollno=int(input("Enter your Roll Number:"))
gpa=float(input("Enter your GPA:"))

print("Name:",name)
print("Roll No:",rollno)
print("GPA:",gpa)
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#01Q1.py
Enter your Name:Moazzam Farooqui
Enter your Roll Number:68
Enter your GPA:3.84
Name: Moazzam Farooqui
Roll No: 68
GPA: 3.84
```

Q2. Create variables of type int, float, string, and bool, then print their values and data types using type()

SOLUTION:

```python
rollno=68
gpa=3.84
name="Moazzam Farooqui"
is_student=True

print(rollno,type(rollno))
print(gpa,type(gpa))
print(name,type(name))
print(is_student,type(is_student))
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#01Q2.py
68 <class 'int'>
3.84 <class 'float'>
Moazzam Farooqui <class 'str'>
True <class 'bool'>
```

Q3. Take a number as input and print its square, cube, and half value.

SOLUTION:

```python
number=float(input("Enter the number:"))

is_square=number*number
is_cube=number*number*number
is_half=number/2

print("Square of the number is:",is_square)
print("Cube of the number is:",is_cube)
print("Half of the number is:",is_half)
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#01Q3.py
Enter the number:5
Square of the number is: 25.0
Cube of the number is: 125.0
Half of the number is: 2.5
```

EXERCISE # 02: CONDITIONS AND LOOPS

Q1. Take a number as input and print its square, cube, and half value.

SOLUTION:

```python
def is_even_odd():
    number=int(input("Enter the number:"))
    if(number%2==0):
        print(f"{number} is an even number")
    else:
        print(f"{number} is an odd number")

is_even_odd()
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01>  python Lab#01Ex#02Q1.py
Enter the number:5
5 is an odd number
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#02Q1.py
Enter the number:6
6 is an even number
```

Q2. Print all numbers from 1 to 20 using a for loop.

SOLUTION:

```python
for number in range(1,21):
    print(number)

```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#02Q2.py
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
```

Q3. Use a while loop to print the multiplication table of a given number.

SOLUTION:

```python
number=int(input("Enter the number: "))

m=1

while m<=10:
    print(f"{number}x{m}={number*m}")
    m+=1
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#02Q3.py
Enter the number: 9
9x1=9
9x2=18
9x3=27
9x4=36
9x5=45
9x6=54
9x7=63
9x8=72
9x9=81
9x10=90
```

Q4. Challenge: Count how many numbers between 1 and 50 are divisible by 3.

SOLUTION:

```python
count=0
for number in range(1,51):
    if number%3==0:
        count+=1

print(f"Numbers divisible by 3: {count}")
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#02Q4.py
Numbers divisible by 3: 16
```

EXERCISE # 03: FUNCTIONS AND DATA STRUCTURES

Q1. Create a function that accepts two numbers and returns the larger one.

SOLUTION:

```
def greater_number():
    number1=int(input("Enter number 1:"))
    number2=int(input("Enter number 2:"))
    if(number1>number2):
        return number1
    else:
        return number2

print(greater_number())
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#03q1.py
Enter number 1:4
Enter number 2:3
4
PS C:\users\user\Desktop\ITTDS Labs\LAB#01>  python Lab#01Ex#03q1.py
Enter number 1:6
Enter number 2:7
7
```

Q2. Store five course names in a list and print them one by one.

SOLUTION:

```
mylist=['DSA','DBMS','AIES','SE','ITTDS']

for courses in mylist:
    print(courses)
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#03Q2.py
DSA
DBMS
AIES
SE
ITTDS
```

Q3. Create a dictionary containing student information such as name, age, department, and semester.

SOLUTION:

```python
mydictionary={
    "Name":"Moazzam Farooqui",
    "Age":19,
    "Department":"CSIT",
    "Semester":"5th"
}

print(mydictionary)
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#03Q3.py
{'Name': 'Moazzam Farooqui', 'Age': 19, 'Department': 'CSIT', 'Semester': '5th'}
```

Q4. Challenge: Write a function that accepts a list of numbers and returns the maximum, minimum, and average.

SOLUTION:

```python
def calculate(numbers):
    maximum=max(numbers)
    minimum=min(numbers)
    average=sum(numbers)/len(numbers)
    return maximum,minimum,average

numbers=[1,2,3,4,5]

maximum,minimum,average=calculate(numbers)

print("Maximum:",maximum)
print("Minimum:",minimum)
print("Average:",average)
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#03Q4.py
Maximum: 5
Minimum: 1
Average: 3.0
```